



V_{MC} = Ratio of available engine power vs Available rudder power
Performance = Increased rate of climb

FACTOR	V_{MC}	CONTROL	PERFORMANCE	CONTROL NOTES	PERFORMANCE NOTES
CG – Forward	Lowers V_{MC} (Good)	Increases	Decreases	More forward CG creates a longer arm between CG & rudder, which equals more leverage, making rudder more effective. So less rudder is needed to counter asymmetric turning	Forward CG requires more downward force on the tail to keep AC level, which adds to the overall vertical downward force, therefore decreasing lift (= decreased performance)
Bank Angle (3 to 5°)	Lowers V_{MC} (Good)	Increases	Increases	Banking toward operating engine turns AC in same direction as required rudder input - so banking results in less rudder input needed	3-5° bank reduces side slip, which reduces drag and increases performance. No bank would cause more drag, and over-banking (> 5°), will cause loss of VLC needed to offset weigh (decreasing performance)
Weight - Increased (when banking)	Lowers V_{MC} (Good)	Increases	Decreases	When turning, more weight requires more lift. When VL is increased, HL is also increased, which works in same direction as rudder - requiring less rudder input	Lighter weight would provide better rate of climb (performance), but would result in increased V_{MC} (see V_{MC} note)
Power - Lower	Lowers V_{MC} (Good)	Increases	Decreases	Lower power creates less asymmetric turning, thereby requiring less rudder input	Lower power decreases climb (less performance)
Density Altitude – High	Lowers V_{MC} (Good)	Increases	Decreases	Higher DA results in less air over wing/prop, creating less turning tendency, but rudder is not substantially affected and remains effective, so control increases. Higher DA decreases adverse yaw effect	Less dense air decreases performance - same as having less available engine power. More dense air would be like having more power, which would require more rudder input
Drag Devices - Extended	Lowers V_{MC} (Good)	Increases	Decreases	Reward flowing airstream of prop is interrupted by gear & flaps. The interrupted airflow (less airflow) creates less asymmetric turning tendency	Additional drag decreases rate of climb
Prop - Feathered	Lowers V_{MC} (Good)	Increases	Increases	Feathered prop reduces yawing toward the dead engine	Feathered prop decreases drag. Less drag results in increased performance